MATH 200. Midterm 1 Cheat Sheet

→ Vectors:

- $\vec{u}\cdot\vec{v} = |\vec{u}||\vec{v}|\cos\theta$; $\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$; $\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$
- $\text{proj}_{\vec{u}}\vec{v} = \left(\frac{\vec{u}\cdot\vec{v}}{|\vec{u}|^2}\right)\vec{u}$
- $|\vec{u}\times\vec{v}| = |\vec{u}||\vec{v}|\sin\theta$; $\hat{i}\times\hat{j} = \hat{k}$ $\hat{j}\times\hat{k} = \hat{i}$ $\hat{k}\times\hat{i} = \hat{j}$
- Cross product = area of llgm spanned by $\vec{u}$ and $\vec{v}$
- $\vec{v}\times\vec{w} = -\vec{w}\times\vec{v}$
- $\vec{v}\times\vec{v} = 0$
- $\vec{v}\times(\vec{w}+\vec{u}) = \vec{v}\times\vec{w} + \vec{v}\times\vec{u}$
- Area of Δ spanned by $\vec{u}$ and $\vec{v}$ = $\frac{1}{2}|\vec{u}\times\vec{v}|$
- Parallelopiped volume = $(\vec{u}\times\vec{v})\cdot\vec{w}$ ($\vec{w}$, $\vec{u}$, $\vec{v}$ span the parallelopiped)
- If $(\vec{AB}\times\vec{AC})\cdot(\vec{AD}) = 0$, A,B,C,D are coplanar.

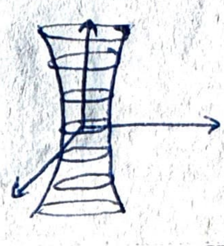
→ Lines and planes

- Line: $L(t) = \vec{P} + t\vec{v}$
- Plane: $\vec{a}\cdot\vec{n} = 0$ ($\vec{a}$ is vector on plane)
- Plane: $\vec{P} + t\vec{v}_1 + s\vec{v}_2$
- line of intersection of planes Π_1 and Π_2: $L = \vec{P} + t(\vec{n}_1\times\vec{n}_2)$
- Distance betwn. point and line: $d = \frac{|\vec{PS}\times\vec{v}|}{|\vec{v}|}$ (S is point, P is point on line)
- Distance betwn. point and plane: $d = \left|\vec{PS}\cdot\frac{\vec{n}}{|\vec{n}|}\right|$

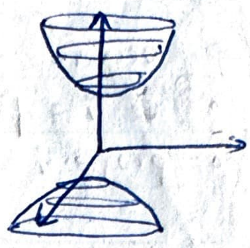
→ Quadric surfaces

- Ellipsoid: $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1$; Based on ellipse: $\frac{x^2}{a} + \frac{y^2}{b} = 1$
- Elliptic paraboloid: $\frac{z}{c} = \frac{x^2}{a} + \frac{y^2}{b}$; Based on parabola: $y = \frac{x^2}{a}$
- Hyperboloid of one sheet: $\frac{x^2}{a} + \frac{y^2}{b} - \frac{z^2}{c} = 1$

If $z = 0$, the graph forms a circle on xy plane.

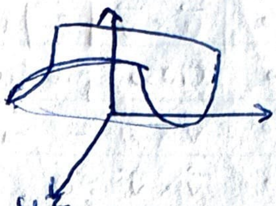


- Hyperboloid of two sheets: $\frac{z^2}{c} - \frac{x^2}{a} - \frac{y^2}{b} = 1$



If $z=0$, the graph does not cut xy plane

- Hyperbolic paraboloid: $\frac{z}{c} = \frac{x^2}{a} - \frac{y^2}{b}$



If $y=0$, $\frac{x^2}{a} = \frac{z}{c}$, concave up parabola in xz
If $x=0$, $-\frac{y^2}{a} = \frac{z}{c}$, concave down parabola in yz

- Double cone: $\frac{z^2}{c} = \frac{x^2}{a} + \frac{y^2}{b}$

→ Vector valued functions $\vec{r}(t)$
Arclength $= \int_{t_1}^{t_2} |\vec{r}'(t)| \, dt$

→ Partial derivatives:

- $f_{xy} = (f_x)_y = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$ $[f_{xy} = f_{yx}]$
- gradient: $\nabla f(x,y) = \langle f_x, f_y \rangle$ → $\perp$ to level sets
following the gradient increases the function the most (steepest)
- linearisation: $L(x,y) = f(a,b) + f_x(x-a) + f_y(y-b)$
- If $\hat{u} = a\hat{i} + b\hat{j}$, $D_{\hat{u}} f = af_x + bf_y = \nabla f \cdot \hat{u} = |\nabla f||\hat{u}|\cos\theta$

→ Double integrals
$$\iint_R f(x,y)\,dx\,dy = \int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_b^a f(x,y)\,dx\,dy$$
Fubini theorem

→ Directional derivatives:
$$D_{\hat{u}} f = a \cdot f_x + b \cdot f_y, \text{ where } \hat{u} \text{ (unit vector)} = \langle a, b \rangle$$
$$= \hat{u} \cdot \nabla f$$
$$= |\nabla f| \, |\hat{u}| \cos\theta$$
$$= |\nabla f| \cos\theta \quad (|\hat{u}| = 1)$$

MATH 200 midterm 2 Cheat Sheet

→ Polar coordinates correction factor: $(r)\,dr\,d\theta$

↔ Spherical coordinates correction factor: $(\rho^2 \sin\varphi)\,d\rho\,d\varphi\,d\theta$

→ Line integrals of scalar functions: finds the area of "shower curtain".

$\int_C f(x,y,z)\,ds$ or $\int_C f(x,y,z)\,dr$ $\quad dr$ or $/ds = \frac{dr}{dt}\cdot dt = |r'(t)|\,dt$

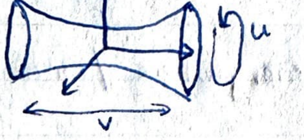
→ Line integrals of vector fields: finds the work done moving over path

$W = \int_C \vec{F}\cdot d\vec{r} = \int_C \underbrace{\vec{F}\cdot r'(t)}_{\text{dot product}}\,dt$

- Parametrisation: parametrise C as $r(t)$, find $\vec{F}(\vec{r}(t))$, then $\int_C \vec{F}(\vec{r}(t))\cdot\vec{r}\,'(t)\,dt = W$
  Bounds of integration: $a \le t \le b$
- Fundamental Theorem of line integrals: If $\vec{F} = \nabla f$, $\int_C \vec{F}\cdot d\vec{r} = f(b) - f(a)$
  If $\text{curl}_2 \vec{F} = 0$, and $\vec{F}$ is defined everywhere $\to \vec{F} = \nabla f \quad \therefore W = f(b) - f(a)$
- Closed loop theorem: If c is a closed loop, $\vec{F} = \nabla f$, $\oint_C \vec{F}\cdot d\vec{r} = 0$
- Green's Theorem: If c is a closed loop, oriented ccw, $\vec{F}$ is nice,
  then $\oint_C \vec{F}\cdot d\vec{r} = \iint_R \underbrace{Q_x - P_y}_{\text{curl}_2 \vec{F}}\,dA$ $\quad dA = dy\,dx$ or $dx\,dy$
  R is region bounded by c

→ Surface parametrisation:

- cylinder $(x^2+y^2=a^2)$: $\vec{r}(u,v) = \langle a\cos u, a\sin u, v\rangle$ $\quad 0 \le u \le 2\pi,\ v \in \mathbb{R}$
- paraboloid $(z = x^2+y^2)$: $\vec{r}(u,v) = \langle u, v, u^2+v^2\rangle$ or $\vec{r}(u,v) = \langle \sqrt{v}\sin u, \sqrt{v}\cos u, v\rangle$
- sphere $(x^2+y^2+z^2=a^2)$: $\vec{r}(\varphi,\theta) = \langle a\sin\varphi\cos\theta, a\sin\varphi\sin\theta, a\cos\varphi\rangle$
- Hyperboloid $(x^2-4y^2+z^2 = 1)$: $x^2+z^2 = 1+4y^2$ (circle in xz with radius $\sqrt{1+4y^2}$)
  $\therefore \vec{r}(u,v) = \langle \sqrt{1+4y^2}\cos u, v, \sqrt{1+4y^2}\sin u\rangle$



- If a surface is within another surface, e.g. $z = x^2+y^2$ with in $x^2+4y^2=9$ (paramenise)
  $\therefore \vec{r}(u,v) = \langle u, v, u^2+v^2\rangle$ subject to restriction: $u^2+4v^2 \le 9$

→ Surface integrals of a scalar function

- Surface area: $\iint_S dS = \iint_{D_{u,v}} |\vec{r}_u \times \vec{r}_v|\,du\,dv$

→ Surface integrals of vector fields: flux integrals

- $\iint_S \vec{F} \cdot d\vec{S} = \iint_S \underbrace{\vec{F} \cdot \hat{n}}_{\text{cross product}} \, dS = \iint_{D_{u,v}} (\vec{F} \cdot \vec{n}) \, |\vec{r}_u \times \vec{r}_v| \, du\,dv$

  $\hat{n}$ ↑ unit normal

- Gradient method: S = level set of g

  $\hat{n} = \pm \frac{\nabla g}{|\nabla g|}$

- $\hat{n}$ is also a vector ⊥ to surface → small chunk of surface is measured by $\vec{r}_u$ and $\vec{r}_v$

$\therefore \vec{n} = \vec{r}_u \times \vec{r}_v$, $\hat{n} = \pm \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}$

$\therefore \iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \pm \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \, dS = \iint_{D_{u,v}} \vec{F} \cdot \pm \frac{\vec{r}_u \times \vec{r}_v}{\cancel{|\vec{r}_u \times \vec{r}_v|}} \cancel{|\vec{r}_u \times \vec{r}_v|} \, du\,dv$

$= \iint_{D_{u,v}} \vec{F}(\vec{r}(u,v)) \cdot \pm \vec{r}_u \times \vec{r}_v \, du\,dv$

→ 3D curl : $\vec{F} = \langle P, Q, R \rangle$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial x & \partial y & \partial z \\ P & Q & R \end{vmatrix}$$

$\text{curl}_3 \vec{F} = \hat{i}(R_y - Q_z) - \hat{j}(R_x - P_z) + \hat{k}(Q_x - P_y)$

$= \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$

If $\text{curl}_3 \vec{F} = \langle 0,0,0 \rangle$ <u>and</u> $\vec{F}$ is defined everywhere / continuous, then $\vec{F} = \nabla f$

→ NOTE: To find the area enclosed by a parametric curve, create a vector field with curl = 1

Then $\int_C \vec{F} \cdot d\vec{r} = \iint_R 1 \, dA$ = Area

MATH 200 Final Exam Cheat Sheet

→ Stokes' theorem: 3D analogue of Green's theorem

- $\oint_C \vec{F}\cdot d\vec{r} = \iint_S \text{curl}_3\vec{F}\cdot d\vec{S} = \iint_S (\nabla\times\vec{F})\cdot\vec{n}\, dS$

The orientation of C and S must follow the right hand rule – fingers curl in the direction of C and thumb points in direction of $\vec{n}$ OR – keep the surface on your left, with head pointing to normal.

- If 2 boundaries to surface:

$$\iint_S \text{curl}_3\vec{F}\cdot d\vec{S} = \oint_{C_1}\vec{F}\cdot d\vec{r} + \oint_{C_2}\vec{F}\cdot d\vec{r}$$

- conditions: $\vec{F}$ must be defined and "nice", $\vec{n}$ and $\vec{c}$ coherent

→ Divergence:

- $\text{Div}(\vec{F}) = \nabla\cdot\vec{F}$; if $\vec{F} = \langle P, Q, R\rangle$, $\text{div}(\vec{F}) = \langle \partial_x, \partial_y, \partial_z\rangle\cdot\langle P, Q, R\rangle$
- $\text{div}(\vec{F}) = P_x + Q_y + R_z$ [If $\vec{n}$ point away, divergence is +ve]
- $\text{div}(\text{curl}\,\vec{F}) = 0$

<u>NOTE</u>: If $\text{curl}_3\vec{F} = 0$ <u>and</u> domain($\vec{F}$) is simply connected, $\vec{F}$ is conservative.

→ Divergence theorem:

$$\iint_S \vec{F}\cdot d\vec{S} = \iiint_E \text{div}(\vec{F})\, dV \qquad [S \text{ must be closed}]$$

- If $\vec{n}$ point outwards, $\text{div}(\vec{F}) > 0$, if $\vec{n}$ converge, $\text{div}(\vec{F}) < 0$.

→ Optimisation: function $f(x,y)$

Setting $f_x = 0$ and $f_y = 0$, we can find the critical points

If $f_{xx}\cdot f_{yy} - [f_{xy}]^2 > 0$, it is a local min or max

If $f_{xx}\cdot f_{yy} - [f_{xy}]^2 < 0$, it is a saddle point

If f_{xx} and $f_{yy} > 0$, it is a local min

If f_{xx} and $f_{yy} < 0$, it is a local max

→ Lagrange multipliers: If max + min + critical don't exist inside the region, they might exist on the boundary.

$x^2+y^2 \leq a$ → region is closed and bounded. ∴ Extreme value theorem implies ∃ an absolute max and min.

$g(x,y) = x^2+y^2$

It does not exist inside region ∴ it must exist on boundary.

∴ Equations:

$$f_x = \lambda \cdot g_x \quad (1)$$
$$f_y = \lambda \cdot g_y \quad (2)$$
$$x^2+y^2 = a^2 \quad (3)$$

Solve for x and y → possibly multiple solutions

<u>NOTE</u>: If e.g. $\vec{r}(u,v) = \langle u-v, \sin(u+v), u+v \rangle$ at $(2,0,0)$

then $u-v = 2$, $u+v=0$, $u=1$, $v=-1$

<u>NOTE</u>: The flux of the curl of $\vec{F}$ through any closed surface is 0.